

titled (document name)', which includes (1-2 sentence list of topics to cover, kept very vague). The full piece should be approximately (n-paragraphs or other unit of length)".

We then take these prompts, and generate outputs with the model we are studying. Ie: Llama 3.2 3b instruct (Dubey et al., 2024). We split these by "\n\n" and the result is that we have 1 million model generations with approximately 10 million paragraphs which we use as our dataset of model paragraph outputs.

## A.2 OUTLINE PREDICTION

For outlines, we take the model generations from before, and pass them to a new model. We tried various open source models for reproducibility, including "openai/gpt-oss-120b", "meta-llama/Meta-Llama-3-8B-Instruct", "meta-llama/Llama-3.3-70B-Instruct", and "Qwen/Qwen2-72B-Instruct".

We found Llama 3 70B (Dubey et al., 2024) had the best results on the tradeoff between precision and brevity.

The prompt we used was as follows:

Return a short, high-level bullet-point outline of the main ideas from the text you are given. Do NOT include any reasoning.

Rules:

- Make as 4-5 bullet points maximum
- Use numbers to enumerate the bullet points
- Aim to capture main ideas of the whole text in the bullet points
- At most 2 short subpoints per point
- Short phrases only (no lengthy sentences)
- Specific to this text (not generic).

## B BASELINES

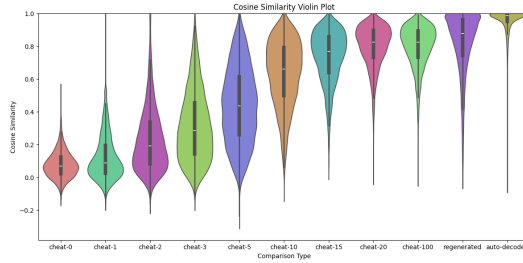


Figure 12: Comparison of how generations starting with some number of cheat tokens compare to regeneration and auto-encoding, using cosine similarity of text embed vectors.

Using the model generated dataset, we compare with the regenerated baseline with the cosine similarity of a text-encoding model (see Appendix C.1). We see that with 10 cheat tokens, the model is already very good at inferring what the model will say, and with 20 cheat tokens there is no substantial difference compared to adding even 100 cheat tokens.

There may be confounders. For example, it has been shown that text-embed models over index on the first few tokens of the embedded text (Lee et al., 2024; Wu et al., 2025). It is also possible that for very short texts, the model may choose to continue the paragraph when it would have counted the paragraph as completed were it to see the text in context. We leave further improvement of this methodology to future work.

## C EVALUATION

### C.1 AUTOMATED METRIC EVALUATION

We first evaluate our methods using established NLP metrics. Using the all-mpnet-base-v2 text embedding model, we compute cosine similarity between generated and reference texts. The SONAR ParaScope methods

(both Linear and MLP variants) achieve mean similarity scores of 0.51 ( $\pm 0.20$ ) and 0.50 ( $\pm 0.20$ ) respectively, significantly outperforming the Continuation ParaScope at 0.33 ( $\pm 0.20$ ). These results position both SONAR methods as comparable to the cheat-5 baseline (0.44), while falling short of full regeneration (0.82).

We supplement these findings with BLEURT scores, which show consistent patterns: SONAR ParaScopes achieve scores of 0.404 ( $\pm 0.144$ ) and 0.395 ( $\pm 0.138$ ), again comparable to the cheat-5 baseline at 0.396 ( $\pm 0.102$ ). The Continuation ParaScope achieves 0.364 ( $\pm 0.135$ ), significantly above random baseline (0.192  $\pm 0.094$ ) but below the regeneration baseline (0.619  $\pm 0.216$ ).

## C.2 SCORING COMPARISON WITH LLM-AS-A-JUDGE

To complement the automated metrics, we performed a fine-grained evaluation using GPT-4o-mini as an evaluator<sup>2</sup>, following the "LLM-as-a-judge" paradigm (Zheng et al., 2023a). We developed a detailed rubric covering multiple aspects of text quality and prompted the LLM to provide brief reasoning before assigning a score for each dimension, a technique shown to improve reliability in rubric-based LLM evaluation (Kim et al., 2024a; Gattani et al., 2024).

We choose 4 main aspects to focus on:

Key dimensions assessed included Coherence (evaluating flow and logical progression, scale 0-3), Subject Match (comparing topic similarity from unrelated to identical focus, scale -1 to 4), Entity Preservation (comparing specific entities mentioned, scale -1 to 4), and Detail Preservation (comparing the specificity of information, scale -1 to 3). The full rubric is provided in Appendix D.

This qualitative assessment revealed distinct trade-offs: the AutoEncoder Map ParaScope methods (Linear and MLP) demonstrated superior preservation of high-level semantic content (Subject and Entities), often matching or exceeding baselines like cheat-5/cheat-10 in topic relevance. Conversely, the Continuation ParaScope consistently generated more coherent and fluent text, scoring higher on the Coherence dimension, though it captured less specific subject matter and fewer entities compared to the reference paragraphs.

## D FULL RUBRIC DETAILS

In this section, we provide the complete evaluation rubric used to assess the quality and similarity of generated paragraphs in our experiments. This rubric was designed to capture multiple dimensions of text quality and semantic similarity between the reference paragraphs and those generated by our ParaScope methods.

### D.1 COMPLEXITY ASSESSMENT

- **0: Trivial** - Text contains minimal content (e.g., only section headers or placeholder text)
- **1: Simple** - Basic content with minimal detail (e.g., simple section headers with brief descriptions)
- **2: Some detail** - Contains short, undetailed sentences about the topic
- **3: Many details** - Contains detailed paragraphs with specific information and nuanced content

### D.2 COHERENCE

- **0: Completely incoherent** - Text contains excessive repetition, nonsensical phrases, or strange symbols
- **1: Partially coherent** - Text is repetitive or has formatting issues (e.g., repeated key phrases, awkward pauses)
- **2: Mostly coherent** - Text has minor errors but maintains logical progression
- **3: Flawless flow** - Text demonstrates logical progression, clear transitions, and no repetition

### D.3 STRUCTURE

- **0: No alignment** - Structural mismatch (e.g., one is a title, the other a paragraph)
- **1: Partial overlap** - Some structural similarities but significant differences
- **2: Highly similar structure** - Matching structural elements and organization

<sup>2</sup>Using 'gpt-4o-mini-2024-07-18' via the OpenAI API

#### D.4 SUBJECT MATCH

- **-1: No subjects to compare** - Insufficient content for comparison
- **0: Completely unrelated subjects** - Topics from entirely different domains (e.g., "corporate law" vs. "particle physics")
- **1: Vaguely similar field** - Subjects from broadly related areas (e.g., "biology" vs. "physics" as sciences)
- **2: Related general domain** - Adjacent fields or related domains (e.g., "history" vs. "archaeology")
- **3: Same subject** - Both discuss the same general topic (e.g., "ancient Mayans")
- **4: Identical focus** - Both analyze the exact same specific topic (e.g., "ancient Mayan architecture")

#### D.5 ENTITIES

- **-1: No entities to compare** - Insufficient entities for comparison
- **0: Completely unrelated** - Entities from different categories (e.g., "Norway" vs. "smartphone")
- **1: Vaguely similar category** - Entities of the same type (e.g., countries, people, cities)
- **2: Similar category** - Entities with categorical similarities (e.g., related countries, similar professions)
- **3: Partial identical entities** - Some matching entities with some differences
- **4: Almost all key entities match** - High degree of entity overlap between texts

#### D.6 DETAILS

- **-1: Neither text has details to compare** - Insufficient details for comparison
- **0: Details differ completely** - No overlap in specific information provided
- **1: Minimal depth** - Basic shared details without specifics
- **2: Moderate depth** - Shared details with some supporting facts
- **3: Highly specific details** - Precise, quantitative, or technical details match

#### D.7 TERMINOLOGY

- **-1: No terminology to compare** - Insufficient terminology for comparison
- **0: No shared terms** - Completely different vocabulary and terminology
- **1: Some overlap** - Partial matching of domain-specific terms
- **2: Domain-specific alignment** - Consistent use of field-appropriate terminology

#### D.8 TONE

- **0: Mismatched** - Different registers or sentiment (e.g., clinical vs. casual, positive vs. negative)
- **1: Consistent** - Similar register, formality level, and sentiment

#### D.9 IDENTICAL ASSESSMENT

- **0: Not identical** - Texts differ in content, even if similar
- **1: Identical** - Texts are essentially the same with only minor variations

This comprehensive rubric allowed us to systematically evaluate the quality of our ParaScope-generated paragraphs across multiple dimensions, providing a nuanced understanding of how well our methods capture and reproduce the planning signals present in the model's residual stream.

## E OUTLINE RUBRIC DETAILS

In this section, we provide the complete outline evaluation rubric used to assess the quality and similarity of generated outlines in our experiments. The rubric was designed to capture the key aspects of outline structure and semantic similarity between the reference outlines and those produced by decoding outline embeddings (SONAR). The embeddings are predictions resulting from a linear probe trained to map normalized residual stream diffs to SONAR TAE embeddings.

#### E.1 COMPLEXITY ASSESSMENT

- **0: Trivial** - Text contains minimal content (e.g., only section headers or placeholder text)
- **1: Simple** - Basic content with minimal detail (e.g., simple section headers with brief descriptions)
- **2: Some detail** - Contains short, undetailed sentences about the topic
- **3: Many details** - Contains detailed paragraphs with specific information and nuanced content

#### E.2 COHERENCE (OUTLINE-LEVEL)

- **0: Completely incoherent** - Excessive repetition, nonsensical phrases, or strange symbols
- **1: Partially coherent** - Repetitive or has formatting issues (e.g., repeated key phrases, awkward pauses)
- **2: Mostly coherent** - Minor grouping or ordering issues, but overall logical
- **3: Clear and coherent** - Logical outline structure with clarity

#### E.3 HIERARCHY / STRUCTURE

- **0: No recognizable hierarchy** - Flat or malformed structure
- **1: Basic levels exist** - Often inconsistent
- **2: Mostly correct hierarchy** - Some mismatches present
- **3: Highly similar structure** - Matches closely with minimal deviations

#### E.4 COVERAGE OF KEY SECTIONS

- **0: Most key sections missing** - Outline 2 omits or replaces core sections
- **1: About half present** - Roughly 50% of major sections covered
- **2: Most sections present** - Minor omissions or regroupings allowed
- **3: Full coverage** - All major sections appear (synonyms/regrouping acceptable)

#### E.5 ORDERING / FLOW

- **0: Largely shuffled** - Illogical or inconsistent ordering
- **1: Partial overlap** - Frequent swaps in order
- **2: Mostly consistent** - Minor order swaps only
- **3: Closely aligned** - Order follows reference outline closely

#### E.6 SUBJECT MATCH

- **-1: No subjects to compare** - Insufficient content for evaluation
- **0: Completely unrelated** - Topics from entirely different domains (e.g., "corporate law" vs. "particle physics")
- **1: Vaguely similar field** - Broad overlap only (e.g., "biology" vs. "physics")
- **2: Related general domain** - Adjacent or related fields (e.g., "history" vs. "archaeology")
- **3: Same subject** - Both discuss the same general topic (e.g., "ancient Mayans")
- **4: Identical focus** - Both focus on the exact same subject (e.g., "ancient Mayan architecture")

#### E.7 ENTITIES / KEY CONCEPTS

- **-1: No entities to compare** - Insufficient key terms/entities
- **0: Completely unrelated** - Entities from unrelated categories
- **1: Same category, little overlap** - Entities belong to same type but differ
- **2: Partial overlap** - Some matching entities or synonyms
- **3: Mostly preserved** - Majority of key entities retained
- **4: Nearly identical** - Almost all entities preserved